Band-1 Hadamard Order 3284 — Miyamoto (H_820 Paley-II); Gap 2052 Still OPEN

H_820/C_822/H_3284 PASS; 3284 Miyamoto-only bonus (NOT in SPK); priority-1 n=3236 seed FAIL; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_3188_Miyamoto`; `2026-09-15_Band1_Hadamard_3076_Miyamoto`; `2026-09-15_Band1_Hadamard_3044_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_3284_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 820 and 3284 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Paley type II seed. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 3284 is a **Miyamoto-only bonus** (**not** in SPK): Paley-I **FAIL** ($q = 3283$ not prime); Paley-II **FAIL** ($q = 1641$ not prime); Kronecker **FAIL** ($3284 = 4 \cdot 821$, 821 not Hadamard-admissible). Priority-1 target $q = 809 \to n = 3236$ was **seed-failed** on this box (nested $H_{808} = \text{Kronecker}(H_{404}, H_2)$ needs H_{100}, which has no SPK/Paley/on-box construction this run). `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 3188 on box. This increment attempts the next preferred Miyamoto target after $q = 797$: **priority-1** $q = 809 \to n = 3236$ **SEED FAIL** (no on-box H_{100} for nested Kronecker/Miyamoto). **Shipped instead: order 3284** ($3284 = 4 \cdot 821$, 821 prime $\equiv 1 \pmod 4$). Seed H_{820} via Paley type II ($q = 409$ prime $\equiv 1 \pmod 4$; order $2(q+1) = 820$). Symmetric Paley conference C_{822} over prime 821. $H_{3284} = \text{Miyamoto}(H_{820}, C_{822})$. Unit tests **28/28 PASS**. SPK next gap remains **2052** (OPEN). Order 3284 is **not** in SPK (Miyamoto-only bonus). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 3284; Miyamoto; Paley type II; Paley conference; constructive gap 2052; Miyamoto-only bonus; Band-1; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in $\{\pm 1\}$ satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{820} constructed and verified (Paley type II, $q = 409$). 2. C_{822} constructed and verified (Paley conference, $q = 821$): $CC^\top = 821I$, zero diagonal. 3. H_{3284} constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 4. SPK: $3284 \notin$ SPK (Miyamoto-only bonus). Paley-I/II/Kronecker paths for 3284 **rejected**. Next constructive gap remains **2052** (OPEN). 5. Priority-1 $n = 3236$ **not constructed** (seed FAIL).

2. Arithmetic and eligibility

Quantity	Value
Target order	$3284 = 4 \cdot 821$
$q = 821$	prime, $821 \equiv 1 \pmod 4 \to$ Miyamoto-eligible
Needed H_{q-1}	H_{820}
Needed conference	order $822 = q + 1$
H_{820} path (used)	Paley type II, $q = 409$ (prime $\equiv 1 \pmod 4$); $2(q+1) = 820$
Conference path	Symmetric Paley over prime 821
Paley-I for 3284	**REJECTED**: $q' = 3283$ not prime
Paley-II for 3284	**REJECTED**: $q' = 1641$ not prime
Kronecker for 3284	**REJECTED**: only 4×821; $821 \not\equiv 0 \pmod 4$ (not admissible)
Priority-1 $n = 3236$	**SEED FAIL**: need H_{808} via Kronecker(404×2) H_{404} Miyamoto $q = 101$ need H_{100} (no on-box construction)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{820}	Paley type II, $q = 409$	0	0.3556	**PASS**
C_{822}	Paley conference, $q = 821$	0	0.5475	**PASS**
H_{3284}	Miyamoto(H_{820}, C_{822})	0	14.6662	**PASS**
$n = 3236$	priority-1 seed triage	—	—	**SEED_FAIL_SKIP**

Primary Miyamoto pipeline wall (incl. unit tests + SPK refresh + 3236 triage) ≈ 16.1616 s. Compressed ± 1 matrices saved as `H820_pm1.npz`, `C822_pm1.npz`, `H3284_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible $\leq 2000 \cup$ on-box prior ships (through 3188/796/…) $\cup$ verified $H_{820} \cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4000). **Miyamoto is not folded into SPK**; order 3284 does **not** enter SPK by classical paths.

Quantity	Value
Known count	885
3284 in SPK	**False** (Miyamoto-only bonus)
820 in SPK	True (Paley II $q = 409$)
3188 in SPK	True (prior Paley I)
3076 in SPK	False (prior Miyamoto bonus)
3044 in SPK	False (prior Miyamoto bonus)
2836 in SPK	False (prior Miyamoto bonus)
Next standard gap	**2052** (OPEN)
Missing first (sample)	2052, 2060, 2076, 2092, 2108, 2148, 2156, 2172, 2212, 2284, 2292, 2300…

5. Gap 2052 (still OPEN; non-finding ONLY)

This ship does **not** fill gap 2052. Prior triage stands: Paley-I/II / Kronecker / Miyamoto-Paley-conf all **FAIL** for order 2052. Claim level for 2052 remains method triage / non-finding ONLY. Non-finding $\neq$ counterexample; Hadamard conjecture remains OPEN. No H_{2052} matrix is claimed.

6. Unit tests (28/28)

Test	Result
H4_sylvester	PASS
H20_paley_I_control	PASS
paley_II_q5_H12	PASS
paley_II_q13_H28	PASS
miyamoto_H20_from_H4_conf6	PASS
paley_I_q11_H12	PASS
q409_is_prime	PASS
q409_equiv_1_mod_4	PASS
H820_paley_II_order_formula	PASS
q821_is_prime	PASS
q821_equiv_1_mod_4	PASS
n3284_equals_4_times_821	PASS
H820_order_formula	PASS
C822_order_formula	PASS
paley_I_H3284_path_rejected	PASS
paley_II_H3284_path_rejected	PASS
kronecker_H3284_path_rejected	PASS
reject_all_ones_H8	PASS
conference_6_paley_control	PASS
miyamoto_eligible_3284_arith	PASS
conference_822_shape_expected	PASS
gap_2052_not_filled_by_this_ship	PASS
3284_miyamoto_only_bonus_arith	PASS
q809_is_prime_equiv_1_mod_4	PASS
n3236_equals_4_times_809	PASS
H808_paley_paths_rejected	PASS
H100_spk_paths_rejected	PASS
paley_II_seed_pattern_control	PASS

7. Non-claims (explicit)

- **Not** the Hadamard conjecture.
- **Not** Clay-closed / finished-ToE / GATE-04 unlocked.
- **Not** a claim that gap 2052 is filled.
- **Not** a construction of order 3236.
- **Not** a claim that 3284 enters SPK (Miyamoto-only bonus).
- No RH / P≠NP / YM / Frankl / NS-solved / AC-solved stamps.

8. Next optional

- Method-attack SPK gap **2052** (GS/SDS / Yamada / Williamson–Turyn lit paths).
- Miyamoto $q = 829 \to n = 3316$ (need H_{828} via Paley-I $q = 827$; note 3316 also Paley-II eligible $q = 1657 \to$ in SPK if constructed that way).
- Deferred: $q = 809 \to n = 3236$ if/when on-box H_{100} (Goethals–Seidel / Williamson) becomes available.
- Track B privacy / Object 1 meter await (GATE-04 remains LOCKED).

9. Reproducibility

```
cd /workspace/experiments/band1/hadamard_3284_miyamoto
python3 verify_hadamard_3284.py
# expects overall_pass=true; max|HH^T-nI|=0 for H_820, C_822, H_3284
```

Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`, `H820_pm1.npz`, `C822_pm1.npz`, `H3284_pm1.npz`.

End of Band-1 certificate. Finite constructions ONLY. GATE-04 LOCKED. clay_closed=false.